Data Analytics & Business Intelligence - Task 3

Advanced SQL Analysis, KPI Modeling & Variance Dashboard

Objective

In Task-2, you learned how to:

- Use SQL JOIN to combine multiple tables
- Calculate KPIs such as Total Sales, Profit Margin
- Create a structured BI dashboard
- Analyze regional and category performance

In **Task-3**, you will step into **performance analysis and decision-support analytics**.

The objective is to:

- Perform advanced SQL queries (CASE, Subqueries, HAVING)
- Conduct trend and variance analysis
- Create calculated KPIs
- Compare current vs previous performance
- Build an executive-level dashboard

This task introduces interns to **management-level business intelligence reporting**.

Why This Task Matters (Industry Context)

In real companies:

- Management wants comparison reports (Month vs Previous Month)
- KPIs must show growth or decline
- Data analysts must explain "why" performance changed
- Dashboards must support strategic decisions

Task-3 prepares interns for **real BI analyst responsibilities**.

What You Will Build

You will build:

- Advanced SQL analysis
- Performance variance calculations
- Growth metrics
- An executive summary dashboard
- A business recommendation report

Dataset Structure (Same as Task-2)

Orders Table

- Order_ID
- Order_Date
- Customer_ID
- Category
- Sales
- Profit
- Discount

Customers Table

- Customer_ID
- Customer_Name
- Region
- Segment

Part A: Advanced SQL Analysis

Step 1: Monthly Performance Analysis

SQL

```sql
SELECT
    YEAR(Order_Date) AS Year,
    MONTH(Order_Date) AS Month,
    SUM(Sales) AS Monthly_Sales,
    SUM(Profit) AS Monthly_Profit
FROM Orders
GROUP BY Year, Month
ORDER BY Year, Month;
```

Step 2: Growth Rate Calculation (Using Subquery)

SQL

```sql
SELECT
    t1.Month,
    t1.Monthly_Sales,
    (t1.Monthly_Sales - t2.Monthly_Sales) / t2.Monthly_Sales * 100 AS Growth_Percentage
FROM
    (SELECT MONTH(Order_Date) AS Month,
            SUM(Sales) AS Monthly_Sales
     FROM Orders
     GROUP BY Month) t1
JOIN
    (SELECT MONTH(Order_Date) AS Month,
            SUM(Sales) AS Monthly_Sales
     FROM Orders
     GROUP BY Month) t2
ON t1.Month = t2.Month + 1;
```

Step 3: Using CASE for Business Classification

Classify orders:

```sql
SELECT
    Order_ID,
    Sales,
    CASE
        WHEN Sales > 1000 THEN 'High Value'
        WHEN Sales BETWEEN 500 AND 1000 THEN 'Medium Value'
        ELSE 'Low Value'
    END AS Order_Type
FROM Orders;
```

Step 4: Identify Underperforming Regions

```sql
SELECT c.Region,
       SUM(o.Profit) AS Total_Profit
FROM Orders o
JOIN Customers c
ON o.Customer_ID = c.Customer_ID
GROUP BY c.Region
HAVING Total_Profit < 10000;
```

Part B: Excel Performance Analysis

Step 5: Create Variance Columns

In Excel:

- Current Month Sales
- Previous Month Sales
- Sales Variance = Current - Previous
- Growth % = (Variance / Previous) * 100

Use formulas:

- IF
- SUMIFS
- INDEX + MATCH

Step 6: Create Performance Comparison Table

Compare:

- Region vs Profit Margin
- Category vs Growth %
- Segment vs Revenue Contribution

Part C: Power BI / Tableau Executive Dashboard

Step 7: Create Calculated KPIs

In Power BI:

- Profit Margin = DIVIDE(SUM(Profit), SUM(Sales))
- Growth % using time intelligence
- Contribution % by category

Step 8: Dashboard Structure

Create executive-level dashboard including:

KPI Cards

- Total Sales
- Total Profit
- Profit Margin
- Growth %
- Total Customers

Visuals

- Line Chart: Monthly Sales Trend
- Column Chart: Region vs Sales
- Bar Chart: Category Profit
- Table: Top Customers
- Donut Chart: Segment Contribution

Step 9: Add Time-Based Filters

Add:

- Year filter
- Month filter
- Region filter
- Category filter

Dashboard must be interactive.

Part D: Business Analysis Report

Answer the following:

1. Is the company growing month-over-month?
2. Which region is underperforming?
3. Which category has highest growth?
4. Are discounts reducing profit?
5. Which customer segment contributes most revenue?
6. What strategic actions should management take?

Prepare a 2–3 page structured report.

Expected Output

At the end of Task-3, you must have:

- Advanced SQL queries
- Growth & variance calculations
- Executive dashboard
- Strategic recommendation report

Key Learning Outcomes

After Task-3, interns will be able to:

- Use CASE and subqueries in SQL
- Perform variance and trend analysis
- Create calculated KPIs
- Build executive dashboards
- Interpret business performance strategically

Deliverables

Mandatory:

1. SQL script file
2. Excel performance analysis file
3. Power BI / Tableau dashboard file
4. Dashboard screenshots
5. Business recommendation report

Optional:

- Data model diagram
- Forecast preview

1. Advanced SQL (CASE, Subqueries, HAVING, Business Queries)

SQL CASE Statement Explained

https://www.youtube.com/watch?v=ELiNcfK3e6c

SQL Subqueries Explained Clearly

https://www.youtube.com/watch?v=7S_tz1z_5bA

SQL HAVING Clause vs WHERE

https://www.youtube.com/watch?v=6XKWWPNkWfE

Advanced SQL for Data Analysis

https://www.youtube.com/watch?v=HXV3zeQKqGY

SQL Window Functions (Optional but Helpful)

https://www.youtube.com/watch?v=YkR7c7YkEJ0

Free SQL Practice Platforms

SQL Practice

https://www.sql-practice.com/

LeetCode SQL Practice
https://leetcode.com/problemset/database/
SQLite Online
https://sqliteonline.com/

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2. Excel (Variance, Growth %, Business Analysis)
Excel Variance Analysis Explained
https://www.youtube.com/watch?v=5Jc5wV6b7VY
Growth Percentage Formula in Excel
https://www.youtube.com/watch?v=2V1E6z8QkU0
Advanced Excel Formulas (SUMIFS, INDEX, MATCH)
https://www.youtube.com/watch?v=ShBTJrdioLo
Excel Business Dashboard Tutorial
https://www.youtube.com/watch?v=K74_FNnlIF8
Official Microsoft Excel Help
https://support.microsoft.com/excel

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3. Power BI (Calculated KPIs, Time Intelligence)
Power BI Desktop Free Download
https://powerbi.microsoft.com/desktop/
Power BI Time Intelligence Explained
https://www.youtube.com/watch?v=AGrl-H87pRU
Creating Growth % and Variance in Power BI
https://www.youtube.com/watch?v=3oC3m-2Vb1k
DAX Basics for Beginners
https://www.youtube.com/watch?v=0a2oHbeSx8E
KPI Dashboard Design in Power BI
https://www.youtube.com/watch?v=Z2zWf9v1nZQ
Official Power BI Documentation
https://learn.microsoft.com/power-bi/

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4. Tableau (Alternative to Power BI)
Tableau Public (Free)
https://public.tableau.com/
Tableau Time-Series Analysis
https://www.youtube.com/watch?v=2P9hE0XjQ2Q
Tableau Calculated Fields Tutorial
https://www.youtube.com/watch?v=7QqKjQ6nFJY
Official Tableau Training
https://www.tableau.com/learn/training

5. KPI & Business Intelligence Concepts
Understanding Business KPIs

https://www.youtube.com/watch?v=Qd8z8QYp4xE

Variance Analysis for Business

https://www.youtube.com/watch?v=V6LQpYxkF8U

Executive Dashboard Best Practices

https://www.youtube.com/watch?v=7zQ6cM8yQmE

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6. Sample Business Datasets

Kaggle Sales & Business Datasets

https://www.kaggle.com/datasets

Google Dataset Search

https://datasetsearch.research.google.com/

Data.gov

https://www.data.gov/

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7. End-to-End BI Project Examples

Advanced SQL + Power BI Project

https://www.youtube.com/watch?v=qfyynHBFOsM

Executive Dashboard Project Walkthrough

https://www.youtube.com/watch?v=GfXFmVfSg5M

Business Intelligence Full Project

https://www.youtube.com/watch?v=Z2zWf9v1nZQ

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8. Community Support

Stack Overflow (SQL, Excel, Power BI)

https://stackoverflow.com/

Power BI Community

https://community.powerbi.com/

Tableau Community

https://community.tableau.com/

Reddit Data Analysis

https://www.reddit.com/r/dataanalysis/

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9. Recommended Learning Plan for Task-3

Day 1

Advanced SQL practice (CASE, HAVING, Subqueries)

Day 2

Excel variance and growth calculations

Day 3

Power BI calculated KPIs

Day 4

Dashboard design and time intelligence

Day 5

Business insight writing and reporting

10. Essentials Checklist for Task-3

You must clearly understand:

- SQL CASE statement
- Subqueries and HAVING
- Variance and growth % calculations
- KPI modeling
- Executive dashboard structure
- Strategic business recommendation writing